

Q1: [Estimated Time: 45 min] [24 * 1 = 24 Marks]

Provide Short answers in 1-2 lines.

- i. What is the conductor in fiber optic cable?
- ii. Provide any three factors that affect the propagation of wireless signals.
- iii. In (access) networks, same cable is used both for downstream and upstream transmissions, however, the data transmission rates of downstream transmission is usually higher than upstream transmission, why?
- iv. Which wireless communication technology supports the highest data rate?
- v. Which physical wired medium supports the highest data rate?
- vi. Differentiate between interference and noise in wireless radio.
- vii. How many simultaneous connections per destination a NAT device with a single WAN-side IP address can support?
- viii. Why is DNS called a “best-effort name-to-address translation” service?
- ix. What action will be performed by the TCP receiver, when an in-order segment with expected sequence number is received, and one other segment has ACK pending.
- x. Does UDP need a mechanism to estimate the RTT between the sender and receiver?
- xi. Discuss the significance of rdt_send() function.
- xii. Provide list of parameters (if any), passed to the rdt_send().
- xiii. List any difference between TCP Reno and TCP Tahoe.
- xiv. What can be the maximum size of an IP datagram?
- xv. In IPv4 datagram header, Provide all valid values of 4-bit “version” field.
- xvi. In IPv4 datagram header, what can be the minimum valid value of 4-bit “header length” field.
- xvii. What is the difference between flooding and broadcasting.
- xviii. In a very small AS using OSPF, is it more efficient to use only one single area or several areas?
- xix. What is semi-permanent TCP connection?
- xx. What does prefix mean (or shows) in BGP advertised route?
- xxi. In routing, what is the Difference between the algorithm complexity and the message complexity.
- xxii. The DHCP protocol is used for assigning IP addresses to clients. At which layer it operates?
- xxiii. How NAT (Network Address Translation) violates end-to-end argument?
- xxiv. Does the TCP protocol ensure that packets are never lost in the network?

Q2: [Estimated Time: 45 min] [6 * 4 = 24 Marks]

- (a) What is protocol encapsulation? Illustrate the process briefly with an example showing headers added across layers. The headers must specify some fields at each layer, having some valid example values.
- (b) Illustrate with the help of a diagram, the problem in two-way handshake in TCP connection establishment.
- (c) Using a simple Telnet example, draw a timing sequence diagram to show how TCP manages sequence numbers and acknowledgments. Host A (client) uses an initial sequence number 42, and Host B (server) uses 78.
- (d) Using 5-bit sequence numbers, what is the maximum size of the sender and receiver windows for each of the following protocols?
 - (i) Go-Back-N
 - (ii) Selective-Repeat
- (e) Explain with the help of a labelled diagram, how explicit congestion notification works. How it notifies the sender and receiver about network congestion?
- (f) A packet (IPV4) has arrived in which the offset value is 100, the value of HLEN is 5 and the value of the total length field is 100. What are the numbers of the first and the last byte? (Show all intermediate calculations).

Q3: [Estimated Time: 45 min] [6 * 4 = 24 Marks]

- (a) What is the role of a load balancer in a datacenter network, and at what protocol layer does it primarily operate?
- (b) In the SDN model, what is the role of the southbound API (e.g., OpenFlow), and what are the three main classes of OpenFlow messages?

- (c) Why is it necessary to divide the global Internet routing system into Autonomous Systems (ASes)? Give two key reasons. Differentiate between the primary goals of an Intra-AS routing protocol (like OSPF) and an Inter-AS routing protocol (like BGP).
- (d) What is the purpose of a "backbone" area in a hierarchical OSPF structure, and what is the specific role of an Area Border Router (ABR)? Explain a scenario that addresses both these questions.
- (e) Trace the path of a datagram and its encapsulating frames as it travels from host A to host B on a different subnet via router R. Specifically, describe what happens to the source and destination MAC and IP addresses at each hop. In this process, how does host A know the MAC address of its first-hop router, R?
- (f) What are the two primary motivations for using VLANs (Virtual LANs) within a large campus or enterprise network? What is a "trunk port" on a switch, and what protocol is used to tag Ethernet frames with a VLAN ID as they traverse between switches?

CLO # 4: Apply Socket Programming (client/server) to solve various real-world problems, including ensuring of data integrity.

Q4: [Estimated Time: 45 min] [2 + 3 + 3 + 6 + 6 + 8 = 28 Marks]

- (a) Why is a MAC address described as "portable," while an IP address is not?
- (b) Let Host A has a datagram of size 7000 bytes, which needs to be sent to Host B. There are three links between Host A and Host B, with each link having Maximum Transmission Unit (MTUs) of size, as shown in Figure 1. You need to calculate, how many numbers of (fragmented) datagrams will be at each link, while transferring from Host A to Host B.
- Number of Datagrams at L1
 - Number of Datagrams at L2
 - Number of Datagrams at L3

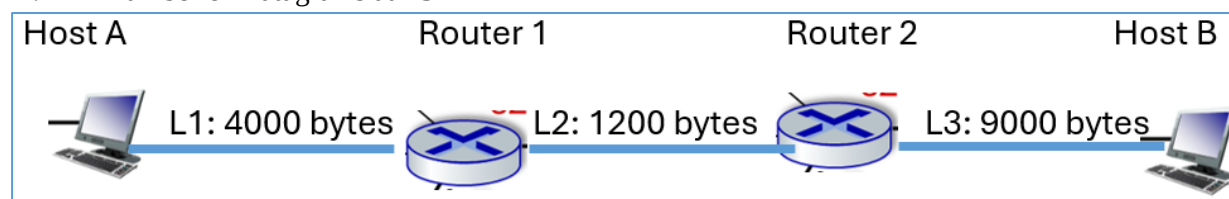


Figure 1. A

Network with MTUs of variable size

- (c) Define route aggregation and explain using an example, how it reduces routing table size.
- (d) Differentiate between RIP, OSPF, and BGP in terms of routing metric and update strategy.

Routing Protocol	Routing metric	Update strategy
RIP		
OSPF		
BGP		

- (e) Provide one advantage and disadvantage for each of the following medium access control protocols.

MAC protocol	Advantage	Disadvantage
Slotted ALOHA		
CSMA/CD		
Polling		

- (f) What field(s) is/are matched for the given devices, and also provide all possible actions for the specified match option.

Device	match	action
Router		
Switch		
Firewall		
NAT		

---The End---